# Sahand Rezaei-Shoshtari

PhD Candidate in Computer Science

| @ sahand.rezaei.sh@gmail.com | sahandrez.github.io | github.com/sahandrez | linkedin.com/in/sahand-rezaei


## Education

**Present – Sep. 2020** | **PhD, School of Computer Science, McGill University and Mila, Montreal, Canada**
**Supervisors:** Doina Precup, David Meger
**CGPA:** 4.00/4.00

**Dec. 2019 – Sep. 2017** | **Master of Engineering - Thesis, McGill University, Montreal, Canada**
**Supervisors:** Inna Sharf, David Meger
**CGPA:** 4.00/4.00
**Thesis:** Learning Manipulator Dynamics for Control and Interaction Inference

**Sep. 2016 – Sep. 2012** | **Bachelor of Mechanical Engineering, University of Tehran, Tehran, Iran**
**CGPA:** 3.98/4.00
**Thesis:** Online Path Planning for a Mobile Robot in Dynamic Environments using Reinforcement Learning

## Work Experience

**Present – Apr. 2024** | **Researcher, Microsoft Research AI for Science, Montreal, Canada**
› AI for science and drug discovery

**Aug. 2023 – May. 2023** | **Research Intern, Microsoft Research AI for Science, Amsterdam, The Netherlands**
› Deep reinforcement learning for chemical reactions discovery

**Feb. 2023 – Apr. 2022** | **Research Intern, Samsung AI Center, Montreal, Canada**
› Hypernetworks for meta imitation learning and meta reinforcement learning

**Sep. 2020 – Mar. 2020** | **Research Intern, Samsung AI Center, Montreal, Canada**
› Multimodal generative modeling for learning intuitive physics
› Deep reinforcement learning for load balancing of 5G networks

**Mar. 2020 – Jan. 2020** | **AI Programmer, Ubisoft La Forge, Montreal, Canada**
› Deep reinforcement learning for automated video game testing

**Aug. 2019 – Mar. 2019** | **Research Intern, Samsung AI Center, Montreal, Canada**
› Object detection neural networks for human-robot interaction

## Publications

2024 **Rezaei-Shoshtari, S.**, Yurchyk, H., Fujimoto, S., Precup, D., and Meger, D., 2024. "Long-Term Fairness in Reinforcement Learning with Bisimulation Metrics". *Under Review*.

2024 Panangaden, P.*, **Rezaei-Shoshtari, S.***, Zhao, R.*, Meger, D., and Precup, D., 2024. "Policy Gradient Methods in the Presence of Symmetries and State Abstractions". *Journal of Machine Learning Research (JMLR)*.

2023 **Rezaei-Shoshtari, S.**, Morissette, C., Hogan, F.R., Dudek, G., and Meger, D., 2023. "Hypernetworks for Zero-shot Transfer in Reinforcement Learning". *The AAAI Conference on Artificial Intelligence*.

2022 **Rezaei-Shoshtari, S.**, Zhao, R., Panangaden, P., Meger, D., and Precup, D., 2022. "Continuous MDP Homomorphisms and Homomorphic Policy Gradient". *Advances in Neural Information Processing Systems (NeurIPS)*.

2021 **Rezaei-Shoshtari, S.**, Hogan, F.R., Jenkin, M., Meger, D., and Dudek, G., 2021. "Learning Intuitive Physics with Multimodal Generative Models". *The AAAI Conference on Artificial Intelligence*.

2021 Hogan, F.R., Jenkin, M., **Rezaei-Shoshtari, S.**, Girdhar, Y., Meger, D., and Dudek, G., 2021. "Seeing Through your Skin: Recognizing Objects with a Novel Visuotactile Sensor". *The IEEE/CVF Winter Conference on Applications of Computer Vision*.

2020 Molamohammadi, M., **Rezaei-Shoshtari, S.**, and Quitoriano, N., 2020. "Jacobian of generative models for sensitivity analysis of photovoltaic device processes". *The Machine Learning for Engineering Workshop at NeurIPS*.

2020 **Rezaei-Shoshtari, S.**, Meger, D., and Sharf, I., 2020. "Learning the Latent Space of Robot Dynamics for Cutting Interaction Inference". *The IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

2019 **Rezaei-Shoshtari, S.**, Meger, D., and Sharf, I., 2019. "Cascaded gaussian processes for data-efficient robot dynamics learning". *The IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

## Skills

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| **Programming** | Python, C++, C#, MATLAB |
| **Machine Learning Frameworks** | PyTorch, TensorFlow, JAX, Hugging Face Libraries, Scikit-Learn |
| **Platforms** | Docker, Slurm Workload Manager, ROS |
| **Other Software** | Git, Mujoco, Unity 3D, OpenCV, PyBullet, MoveIt |

## Select Projects

### LLM Finetuning with Reinforcement Learning from Unpaired Human Feedback — 2023

Ongoing project - Code is not released yet

A novel perspective for finetuning LLMs with unpaired preference data.

RLHF | LLM | PyTorch | Hugging Face Libraries | Python

### Hypernetworks for Zero-Shot Transfer in Reinforcement Learning — 2023

Paper | Webpage | github.com/sahandrez/hyperzero

A novel algorithm based on hypernetworks for zero-shot transfer in deep reinforcement learning.

RL | Hypernetworks | Transfer Learning | PyTorch | Python

### Contextual Control Suite — 2023

github.com/sahandrez/contextual-control-suite

A series of contextual Markov decision processes (MDP) based on Mujoco for continuous control tasks.

RL | Mujoco | Gym | Python

### Deep Homomorphic Policy Gradient (DHPG) Algorithm — 2022

Paper | Webpage | github.com/sahandrez/homomorphic_policy_gradient

A novel algorithm for state-action abstraction and representation learning in deep reinforcement learning.

RL | Representation Learning | PyTorch | Python

### Multimodal Generative Modeling and Visuotactile Simulation — 2020

Paper | Webpage | github.com/sahandrez/multimodal-dynamics

A novel algorithm for multimodal generative modeling of intuitive physics.

Generative Modeling | Multimodal | PyTorch | PyBullet | Python

### Control and Motion Planning Utilities for Kinova Jaco 2 Robot — 2019

github.com/sahandrez/jaco_control

A ROS package for the Kinova Jaco 2 robot with various control and motion planning utilities.

Robotics | ROS | Python

## Select Honors and Awards

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| 2022 – 2025 | Canada Graduate Scholarship-Doctoral (CGS-D) Award. Amount of **$105,000**. NSERC. |
| 2022 – 2026 | Fonds de Recherche du Quebec - Nature et Technologies (FRQ-NT) Award. Amount of **$70,000**. FRQ. |
| 2022 | Grad Excellence Award. Amount of **$7,000** per year. McGill University. |
| 2022 | AAAI Student Scholarship Program. AAAI 2023. |
| 2020 | DeepMind Grad Award. Amount of **$25,000**. DeepMind and McGill University. |
| 2019 | IROS Student and Developing Countries (SDC) Travel Award. IEEE/RSJ IROS 2019. |
| 2017 | Grad Excellence Award. Amount of **$5,000**. McGill University. |
| 2012 | National University Entrance Exam. Ranked **19$^{th}$**. Iran. |

## Community Involvement

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| 2024 – Present | Senior reviewer at RLC. |
| 2022 – Present | Reviewer at NeurIPS, ICML, ICLR. |
| 2022 | NeurIPS 2022 Outstanding Reviewer (Top 8%). NeurIPS. |
| 2022 | ICML 2022 Outstanding Reviewer (Top 10%). ICML. |

## Certifications

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| Apr. 2023 | **Trustworthy and Responsible AI Learning (TRAIL)**, Mila, Montreal, Canada |
| Oct. 2020 | **Simons Institute Mathematics of Online Decision Making Workshop**, Virtual |
| Aug. 2019 | **CIFAR Deep Learning and Reinforcement Learning (DLRL) Summer School**, Edmonton, Canada |